

LANDSLIDE DETECTION MODEL

EARLY WARNING SYSTEM



Prepared by: Komal Niwane
GIS Intern, AIQ Space Ventures, Mumbai

1. What This Model Will Do

The goal is a two-part system for Pune district that tells us both WHERE landslides are likely to occur, and — during the monsoon season — gives an early alert of roughly one week before conditions become critical at those locations, so field/response teams have time to act.

- Susceptibility Layer — a district-wide map showing which slopes are structurally/geologically prone to landslides, built once and updated periodically (not tied to any single storm event).
- Early-Warning Trigger Layer — a monsoon-season monitoring layer that tracks rainfall forecasts and ground-saturation trends, and flags when a susceptible zone is approaching a critical threshold — this is the ~1-week-ahead alert component.
- Combined Output — the two layers are combined into a single, regularly-updated Risk Map with Low / Moderate / High / Very High zones, plus a list of the specific villages/talukas that need attention that week.

2.OVERALL WORKFLOW

Step 1 — Data Collection

Gather terrain, geology, soil, land-cover, historical landslide records, and rainfall data for Pune district (one-time + periodic refresh).

Step 2 — Data Preparation

Bring all data layers onto the same map grid and coordinate system so they can be combined pixel-by-pixel.

Step 3 — Susceptibility Modeling

Train a machine-learning model on historical landslide locations to score every location in the district by structural risk (Low → Very High).

Step 4 — Monsoon Trigger Monitoring

During June–September, ingest the rolling 7-day rainfall forecast and recent rainfall/vegetation trends for the district.

Step 5 — Risk Fusion & Alerting

Combine the static susceptibility score with the current trigger trend to produce a weekly (or more frequent) updated risk status per zone.

Step 6 — Output & Reporting

Publish the updated Risk Map and a short list of watch-zones to stakeholders/response teams.

Step 7 — Post-Season Review

After each monsoon, compare predicted watch-zones against what actually happened, and use that to recalibrate thresholds for the next season.

3. PARAMETERS REQUIRED

3.1 Static Parameters (the "Where")

Parameter	What It Captures	Category
Elevation	Base terrain height	Terrain
Slope	Steepness — the single strongest terrain driver of landslides	Terrain
Aspect	Slope-facing direction (affects sun/moisture exposure)	Terrain
Curvature (profile & plan)	Whether a slope concentrates or disperses water flow	Terrain
Terrain Ruggedness	General surface roughness/instability indicator	Terrain
Topographic Wetness Index	How much a location naturally accumulates water	Hydrology
Stream Power Index	Erosive force of water flow at a location	Hydrology
Distance to drainage/streams	Proximity to natural water channels	Hydrology
Distance to geological faults/lineaments	Proximity to structurally weak zones	Geology
Lithology (rock type)	Rock strength/permeability class	Geology
Soil type & depth	How well the ground holds together and drains	Soil
Land use / land cover	Forest vs. bare vs. built-up — vegetation root cover stabilizes slopes	Land Cover
Distance to roads	Road-cut slopes are a documented destabilization factor	Human Impact
Historical landslide inventory	The actual past event locations — this is what the model learns from	Ground Truth

3.2 Dynamic Parameters (the "When" — Early Warning Layer)

Parameter	What It Captures	Update Frequency
7-day rainfall forecast (IMD)	Forecasted rainfall driving the 1-week lead time	Daily, rolling 7-day window
Antecedent Precipitation Index (API)	How saturated the ground already is from recent actual rainfall	Daily
Vegetation trend (NDVI change)	Detects slope destabilization/deforestation before it's visually obvious	Every 5–10 days (satellite revisit)
Soil moisture trend	Direct/proxy measure of ground saturation	Weekly
Satellite ground-deformation trend (InSAR, if available)	Very early physical slope movement	Every 6–12 days (satellite revisit)

Python Implementation Pipeline:

1. **config.py** — Central configuration — paths, CRS, thresholds, all tunable parameters.
2. **utils_raster.py** — Raster I/O and terrain-derivative helpers (slope, aspect, curvature, flow accumulation, TWI, SPI, distance rasters) — implemented in pure numpy.
3. **feature_engineering.py** — Builds the static feature stack and the Antecedent Precipitation Index (API) time series.
4. **sampling.py** — Extracts training points from the landslide inventory and generates pseudo-absence (negative class) points.
5. **spatial_cv.py** — Spatial block cross-validation, to avoid the optimistic bias of plain random k-fold splits on spatial data.
6. **train_model.py** — Random Forest / XGBoost training, SMOTE class-imbalance handling, and recall-first threshold optimization.
7. **early_warning.py** — Fuses static susceptibility with dynamic rainfall-trigger conditions into a 1-week-ahead risk level.
8. **validate_historical.py** — Backtests the early-warning logic against a known historical landslide event.
9. **generate_synthetic_data.py** — Generates synthetic demo data so the single-district pipeline can be run and verified end-to-end without real data on hand.
10. **main.py** — Orchestrates the full SINGLE-DISTRICT (Pune) pipeline end-to-end, from data loading through susceptibility mapping and historical backtest.
11. **geo_utils.py** — Picks the correct local UTM zone for any latitude/longitude in the country, so multi-region features are in consistent metric units.
12. **multi_site_data.py** — Generates synthetic demo data across SEVERAL regions (Western Ghats, Kerala, Uttarakhand, Sikkim) for the nationwide pipeline.
13. **multi_site_pipeline.py** — Builds a training table across multiple regions and runs leave-one-region-out cross-validation -- the honest test of whether the model generalizes to a region it never saw.
14. **main_national.py** — Orchestrates the NATIONWIDE / multi-region pipeline end-to-end: any location in the country, not just one fixed district AOI.
15. **requirements.txt** — Python package dependencies.
16. **README.md** — How to run both pipelines, switch to real data (district and nationwide), and known limitations.